

PRATHAM AGARWAL

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EDUCATION

Shri Ramdeobaba College of Engineering and Management (CSE-AIML)

2022-2026

EXPERIENCE

Software Engineer Intern - Soroban

Feb 2025-Present

- Developed an API endpoint for theorem explanations by taking user input. Integrated it into the video-gen repository.
- Implemented Git pre-commit and pre-push hooks to enforce branch protection and maintain production code integrity.
- Explored and integrated function-plot and other graphics libraries without affecting the main production environment.
- Conducted competitive analysis and market research on AI-powered EdTech solutions to identify key differentiators, user needs, and pricing models, shaping the product roadmap and strategic positioning.

PROJECTS

Smart Fitness Trainer: Repetition Tracker and Posture Assistant (Team Project) [Link](#)

- Collaborated in developing an AI-powered real-time virtual fitness assistant that utilizes OpenCV and machine learning techniques to track exercise repetitions and detect improper posture.
- Worked on integrating a Flask-based backend to process video frames efficiently, ensuring real-time feedback with response times consistently under 100 milliseconds.
- Assisted in training and fine-tuning convolutional neural networks (CNNs) to improve posture detection accuracy to 95%, enabling precise identification of incorrect movements.
- Contributed to developing an intuitive and user-friendly graphical interface, ensuring a smooth interaction between users and AI models.

Phishing Website Detection Using Machine Learning (Team Project) [Link](#)

- Part of a team that developed a machine learning-based phishing detection model that leverages URL-based and content-based features, achieving an accuracy of 85+%.
- Assisted in applying advanced text processing techniques, such as TF-IDF vectorization, to extract key features from website content for better phishing detection.
- Helped in developing a user-friendly interface for seamless interaction with the phishing detection system.
- Currently contributing to designing a Flask API to serve the trained phishing detection model for real-time threat identification, with Google's Safe Browsing API integration to incorporate double-checking.

TECHNICAL & SOFT SKILLS

- Programming Languages: Python, Java, JavaScript, SQL, C
- Core Concepts: Data Structures, Algorithms, Object-Oriented Programming (OOP)
- Machine Learning: NumPy, Pandas, Matplotlib, Scikit-learn
- Backend Development: Flask, Node.js
- Tools & Platforms: Git, GitHub, Linux, Bash, Docker, Kubernetes, Jupyter Notebook
- Soft Skills: Adaptability, Team Collaboration, Communication Skills, Problem-Solving, Time Management

CERTIFICATES

Introduction to Machine Learning – NPTEL-IITKGP, Oct 2024 [Link](#)

Git and GitHub Bootcamp – Udemy, September 2024 [Link](#)

Introduction to Cloud Computing – edX(IBM), May 2024 [Link](#)

